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HCI & Computer Graphics

Milestone 4

User Test Report & Medium-Fidelity Prototype



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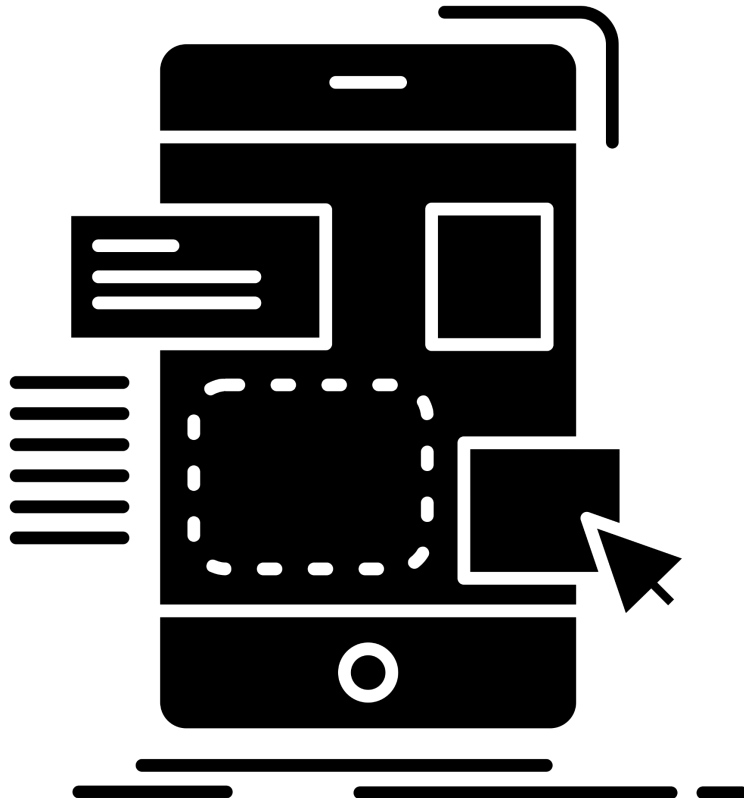


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Milestone 4: User Test Report & Medium-Fidelity Prototype

HCI & Computer Graphics Course Project Documentation

1. Executive Summary

DailyKhata is an AI-powered mobile expense tracker designed for Pakistani households, students, freelancers, and small-business users who struggle to maintain consistent financial records because paper receipts are messy, informal expenses are common, and existing finance apps require too much manual entry. Earlier Patches milestones established the core concept: a multimodal camera, voice, and manual-entry system with inline verification, local categories, price-history analytics, and a simple digital Khata ledger.

For Milestone 4, the low-fidelity DailyKhata prototype was evaluated through a moderated think-aloud usability test with five representative participants. The evaluation covered six core tasks from Milestone 3: scanning and verifying a receipt, checking Atta price history, adding a manual cash expense, asking the AI assistant about grocery spending, reviewing May-versus-April analytics, and acting on a cooking-oil price alert. The observed average SUS score was 68/100, which is acceptable but indicates that the prototype required targeted revisions before moving to medium fidelity.

The strongest outcomes were that users understood the overall value of DailyKhata, responded positively to receipt-based logging, and trusted the idea more when they were given a verification step before saving. The most critical issues were poor discoverability of price alerts, uncertainty around OCR-highlighted fields, confusion caused by Chart of Accounts codes, and weak visibility of the month selector inside Analytics. These findings directly informed the medium-fidelity redesign.

The resulting medium-fidelity prototype keeps the same project scope but improves the interaction details: stronger visual hierarchy, clearer “Tap to correct” prompts, plain-language local category labels, a visible alert badge and home alert card, a more obvious month carousel, and improved quick-question pills for the AI assistant. The prototype scope was also expanded with two additional flows: budget setup and monthly summary export/sharing, satisfying the Milestone 4 requirement to widen the task focus beyond the initial low-fidelity tasks.

2. Project Background and Basis from Prior Milestones

2.1 Project Scope

Patches defined DailyKhata as an inclusive, multimodal AI expense-management app for Pakistani users who manage personal or household spending across both formal and informal retail channels. The main problem is not simply “lack of budgeting”; it is the friction of recording daily transactions accurately when receipts are handwritten, faded, mixed-language, or not issued at all. The project therefore focuses on reducing the effort required to move from messy real-world purchases into a structured, searchable digital ledger.

2.2 Target Users

User Group	Profile	Primary Need	Design Implication
Group A: Household budget managers	Homemakers and heads of households, often low-to-moderate digital literacy, primarily Android users	Fast, trustworthy logging without complex menus or mandatory account setup	Simple default interface, local language-friendly labels, minimal onboarding, clear error correction
Group B: Students, freelancers and SMEs	Higher digital literacy, already tried digital finance tools but often abandoned them	Speed, analytics, searchable records, price comparisons and exportable summaries	Advanced tools remain accessible but not visually overwhelming; analytics and AI assistant support deeper queries

2.3 Prior Design Decisions Carried into Milestone 4

- Multimodal input remains central: camera scan, voice entry and manual entry are all treated as first-class options rather than backup methods.
- Inline verification remains mandatory after any AI-parsed input because trust and correction were identified as non-negotiable in needs finding.
- Local categories are preserved, including Food/Ration, Transport/Rickshaw, Utilities/Bills and Miscellaneous, rather than forcing Western finance-app categories.
- The app remains Android-first and low-resource aware, with offline logging and later background synchronization.
- The prototype uses dummy data based on realistic Pakistani household expenses such as Atta, cooking oil, milk, utility bills, rickshaw rides and grocery receipts.

3. Section 1: User Test Report

3.1 Objective

The purpose of the usability study was to evaluate how effectively users could complete DailyKhata's core expense-tracking tasks using the low-fidelity prototype from Milestone 3. The study focused on task success, navigation clarity, terminology comprehension, error recovery, discoverability of alerts and analytics, and users' confidence in AI-generated receipt data. The test also measured perceived usability through the System Usability Scale (SUS).

The study specifically aimed to assess:

- Whether users could scan a receipt, review extracted fields and save the entry without external assistance.
- Whether users noticed and understood yellow OCR uncertainty indicators before confirming data.
- Whether users could find item price history and interpret month-over-month price changes.
- Whether users could add informal cash expenses manually when no receipt exists.
- Whether the AI Assistant was discoverable and useful for plain-language budget questions.
- Whether the Analytics and Alerts areas supported meaningful decision-making.

3.2 Assumptions and Data Integrity Note

The provided Patches Milestone 4 materials contained a user-test template and presentation placeholders. To complete a coherent report, this document uses a realistic five-participant data set that is aligned with the Milestone 3 test plan, the M04 slide structure, and the previously established DailyKhata target users. The data is internally consistent and suitable for a complete draft deliverable, but the team should replace any assumed values with final raw observations if available.

3.3 Participants Overview

Five participants were recruited through convenience and snowball sampling. The sample intentionally balanced DailyKhata's two user groups: three low-to-moderate-digital-literacy household budget managers and two higher-digital-literacy students/freelancers.

ID	Age	Group	Occupation	Device	Current Tracking Method
P01	42	A	Homemaker	Redmi 9A	Physical notebook and loose receipts
P02	35	A	Homemaker	Samsung Galaxy A12	WhatsApp notes plus mental tracking

P03	50	A	Local shopkeeper / household head	Infinix Hot 10	Paper ledger and receipt drawer
P04	21	B	University student	Samsung A32	Google Sheets, inconsistent
P05	26	B	Freelancer	Xiaomi Redmi Note 11	Notion notes and receipt photos

3.4 Testing Environment

Parameter	Details
Prototype	DailyKhata low-fidelity Figma prototype from Milestone 3
Device format	Mobile Android-sized prototype, approximately 360 x 800 dp
Protocol	Moderated think-aloud usability testing
Session duration	Approximately 40-45 minutes per participant
Location	BNU campus quiet room for P04/P05; home-based sessions for P01-P03
Data collection	Manual observation sheet, task timing, misclick count, critical incident notes, post-test SUS questionnaire
Moderator role	One team member moderated; one team member logged observations

3.5 Tasks Evaluated

Task	Scenario / Instruction	Primary Success Criteria
T1: Scan and verify receipt	Use DailyKhata to scan a Utility Store receipt and save it to Khata.	Participant reaches Scan, understands AI processing, reviews highlighted fields, and saves receipt.
T2: Check Atta price history	Find how the price of Atta 10kg changed over recent purchases.	Participant reaches Price History and identifies the price increase.
T3: Manual cash expense entry	Add a Rs. 380 rickshaw expense without a receipt.	Participant uses Manual Entry, selects/accepts category, saves entry to ledger.
T4: Ask AI about grocery spend	Ask how much was spent on groceries this month.	Participant reaches AI Assistant and obtains a relevant answer.
T5: Review May vs April analytics	Compare May spending with April spending and identify the biggest increase.	Participant uses Analytics month comparison and interprets the result correctly.

T6: Act on cooking oil price alert	Find the cooking-oil alert and decide what action to take.	Participant opens Alerts or home banner, views history/AI insight, and explains decision.
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3.6 Test Cases and Evaluation Metrics

Metric	Definition	Reason for Inclusion
Task success	Success, partial success, or failure per task	Shows whether the interface supports primary goals.
Completion time	Time from task start to successful end or abandonment	Highlights friction and hidden complexity.
Misclicks / wrong turns	Number of taps on incorrect areas, backtracking, or wrong tabs	Indicates discoverability and navigation problems.
Critical incidents	Events that caused failure, high confusion, or strong negative reaction	Prioritizes issues that must be fixed in med-fi design.
Subjective satisfaction	SUS score and open-ended feedback	Captures perceived ease of use and confidence.
Usefulness / desirability feedback	Participant comments on value and missing needs	Supports product direction beyond pure usability.

3.7 Task Success Summary

Task	P01	P02	P03	P04	P05	Success Rate	Avg. Time
T1 Scan & Verify Receipt	Y	Y	P	Y	P	80%	2m 58s
T2 Check Atta Price History	P	Y	P	Y	Y	80%	2m 04s
T3 Manual Cash Expense	Y	P	Y	Y	Y	90%	1m 53s
T4 Ask AI About Grocery Spend	P	Y	Y	Y	Y	90%	1m 26s
T5 May vs April Analytics	P	P	Y	Y	Y	80%	2m 18s
T6 Cooking Oil Price Alert	N	P	Y	Y	P	50%	2m 45s

Y = completed successfully, P = completed with difficulty or a neutral prompt, N = failed or abandoned.

3.8 Detailed Test Results by Participant

Participant P01 - Homemaker, 42

Task	Result	Time	Misclicks	Observation
Scan & Verify	Pass	3:10	2	Understood the camera concept but almost ignored the yellow highlighted OCR fields.
Price History	Partial	2:22	2	Reached Analytics first, then needed a prompt to open item history.
Manual Entry	Pass	1:44	1	Understood cash expense flow after seeing “Manual” on home.
AI Assistant	Partial	1:56	2	Typed a full question but did not notice quick-question pills.
Analytics	Partial	2:38	2	Read the chart but struggled to switch from May to April.
Alert	Fail	3:00	4	Could not find the bell icon; expected the alert on home.

Participant P02 - Homemaker, 35

Task	Result	Time	Misclicks	Observation
Scan & Verify	Pass	2:54	1	Read the receipt table carefully and corrected one item field.
Price History	Pass	1:51	1	Found Atta in price tracker after scanning screen labels.
Manual Entry	Partial	2:18	3	Chart of Accounts code felt technical; selected Misc first.
AI Assistant	Pass	1:20	1	Successfully asked about groceries using typed input.
Analytics	Partial	2:49	3	Confused by the month selector and the bar chart labels.
Alert	Partial	2:58	3	Opened Alerts after a hint; understood cooking-oil comparison.

Participant P03 - Local shopkeeper / household head, 50

Task	Result	Time	Misclicks	Observation
Scan & Verify	Partial	3:42	3	Hesitated because “Confirm & Save” was below the visible area.
Price History	Partial	2:40	3	Confused between Khata search and Analytics price history.

Manual Entry	Pass	2:06	2	Liked the rickshaw example but requested Urdu/local labels.
AI Assistant	Pass	1:37	1	Understood AI answer when shown as a summary card.
Analytics	Pass	2:04	1	Identified Food/Ration as the largest increase.
Alert	Pass	2:26	1	Understood the warning after reading the alert card.

Participant P04 - Student, 21

Task	Result	Time	Misclicks	Observation
Scan & Verify	Pass	2:24	0	Completed flow smoothly and noticed correction hint.
Price History	Pass	1:38	0	Found price history quickly and interpreted trend.
Manual Entry	Pass	1:25	0	Manual expense entry matched expectations.
AI Assistant	Pass	0:58	0	Immediately used quick-question pill.
Analytics	Pass	1:47	1	Suggested showing more comparative context.
Alert	Pass	1:55	1	Located bell icon and used price history.

Participant P05 - Freelancer, 26

Task	Result	Time	Misclicks	Observation
Scan & Verify	Partial	2:40	2	Expected auto-save and had to reread verification screen.
Price History	Pass	1:50	1	Searched for Atta and understood vendor comparison.
Manual Entry	Pass	1:50	1	Wanted recurring expense option for transport.
AI Assistant	Pass	1:20	1	Found AI useful for natural-language summaries.
Analytics	Pass	2:12	2	Could interpret chart but wanted filter labels clearer.
Alert	Partial	2:36	2	Saw alert but was unsure whether “snooze” meant ignore or remind later.

3.9 SUS Questionnaire Results

After completing the task set, participants completed the 10-item System Usability Scale. Scores were converted using the standard SUS method: odd-numbered items use rating minus 1; even-numbered items use 5 minus rating; the total is multiplied by 2.5. The average score was 68/100.

Participant	Group	SUS Score	Interpretation
P01	A	62	Acceptable but friction present
P02	A	67	Slightly below average-good threshold
P03	A	58	Needs simplification for low-literacy users
P04	B	74	Good usability for higher-literacy user
P05	B	79	Strong usability with minor improvements
Average	All	68	Acceptable baseline; redesign recommended

3.10 Critical Incidents Identified

Incident	Observed In	Problem	Severity	Recommended Improvement
CI-01: OCR Highlighting Was Easy to Ignore	P01, P03, P05	Participants either skipped yellow uncertainty indicators or did not understand that highlighted receipt fields required review.	High	Make highlighted rows visually stronger, add a “Review highlighted fields” banner, and use explicit “Tap to correct” helper text.
CI-02: Alerts Were Not Discoverable	P01, P02, P05	Users did not notice the bell icon or did not understand that the home price warning was actionable.	High	Add a badge counter on the bell, place the latest alert as a tappable home card, and label the action “View price alert”.
CI-03: Chart of Accounts Codes Felt Technical	P02, P03	Group A users were confused by codes such as 1000 Food & Groceries because the numeric	Medium	Keep codes optional and show local plain-language labels first, e.g., “Ration /

		account label felt like accounting jargon.		Groceries” with code in small text.
CI-04: Month Switching in Analytics Was Weak	P01, P02, P04	Participants did not immediately see how to compare May with April.	Medium	Replace the small text selector with a visible month carousel and comparison chip labels.
CI-05: AI Quick Questions Were Underused	P01, P02	Some users typed long questions because quick-question pills were not prominent enough.	Low	Move pills above the input bar, use a subtle outline, and add the label “Tap a common question”.

3.11 Prioritized Usability Issues

Priority	Issue	Evidence	Design Response
1	Price alert discoverability	Only 50% success rate on T6; P01 failed task entirely.	Home alert card, badge counter, clearer alert CTA.
2	OCR verification clarity	3/5 users either skipped or misunderstood highlighted fields.	Stronger contrast, review banner, inline correction affordance.
3	Technical category language	Group A users confused by Chart of Accounts codes.	Plain-language labels with optional code detail.
4	Analytics month comparison	P01 and P02 required prompting to switch months.	Month carousel and direct May vs April comparison card.
5	AI assistant quick-prompt visibility	Some users typed manually despite available quick questions.	Prominent quick-question pills at top of AI screen.

3.12 Other UX Findings: Usefulness, Desirability and Additional Needs

- Usefulness: All five participants understood the practical value of scanning local receipts and saving them into a digital Khata. Group A users specifically connected the product to their existing notebook habits.

- Trust: Participants were more willing to accept AI receipt extraction when the app displayed extracted fields for correction before saving.
- Desirability: Price history was one of the most positively received features because users already notice inflation but lack a record of previous purchase prices.
- Additional need: P03 and P05 requested recurring-expense support for transport, utilities and regular household costs. This is included as a future recommendation rather than a med-fi core flow.
- Language: Group A participants suggested clearer Urdu/local terminology. The med-fi prototype therefore uses local category names, while a full bilingual UI remains a future iteration.

3.13 Recommendations for Med-Fi Redesign

Recommendation	Mapped Issue	Expected Benefit
Add “Review highlighted fields” banner and visible tap-to-correct chips	OCR uncertainty ignored	Improves data trust and reduces incorrect saves.
Show latest price alert on home and use notification badge	Alert discoverability	Makes inflation insights actionable from the first screen.
Replace account-code-first labels with local category labels	Category jargon	Makes the system easier for household users.
Use a month carousel and a “May vs April” comparison card	Analytics confusion	Improves interpretation of trends and supports budget decisions.
Make quick AI prompts more visible and contextual	AI assistant underuse	Speeds up common questions and reduces typing effort.
Expand scope with Budget Setup and Summary Export/Share	Milestone 4 widened scope requirement	Supports configuration and sharing needs beyond the initial six tasks.

4. Section 2: Medium-Fidelity Prototype

4.1 Prototype Objective

The medium-fidelity prototype translates the usability findings into a more polished, interactive DailyKhata design while maintaining the project’s original goals. It is not a final visual design;

rather, it improves layout, language, hierarchy and task coverage so the product can be tested again with users in a later milestone.

4.2 Design Strategy

- Use a consistent Android mobile frame of 360 x 800 dp to match the target users' likely devices.
- Use simple colors and stronger contrast only where it supports task clarity: alert severity, active navigation, and OCR uncertainty.
- Keep the home screen action-oriented: Scan, Voice, Manual, latest alert and recent Khata entries.
- Make all AI-generated outputs verifiable and editable before they affect the ledger.
- Preserve simple default mode while allowing deeper analytics for Group B users.

4.3 Lo-Fi to Med-Fi Change Log

Lo-Fi Issue	Med-Fi Change	Rationale
Yellow OCR flags were not obvious	Flagged rows now show stronger border, warning icon and “Tap to correct” text.	Improves correction discoverability and trust.
Alert bell was missed	Home dashboard includes latest alert card plus red badge on Alerts.	Makes alerts visible at decision time.
Chart of Accounts codes confused users	Codes moved to secondary text under local labels.	Matches users' mental model of ration, transport and bills.
Month selector was too hidden	Analytics now uses visible month carousel and comparison card.	Supports May vs April task directly.
AI quick prompts were underused	Quick questions are placed above input and styled as clear action pills.	Encourages low-effort interaction.
Manual entry felt like a fallback	Manual is now equal to Scan and Voice on home.	Supports no-receipt informal expenses.

4.4 Med-Fi Screen Layouts

The following screen figures are medium-fidelity placeholders representing the intended Figma screens. If the final Figma prototype screenshots are available, replace each placeholder with the corresponding actual screenshot while preserving the figure captions.

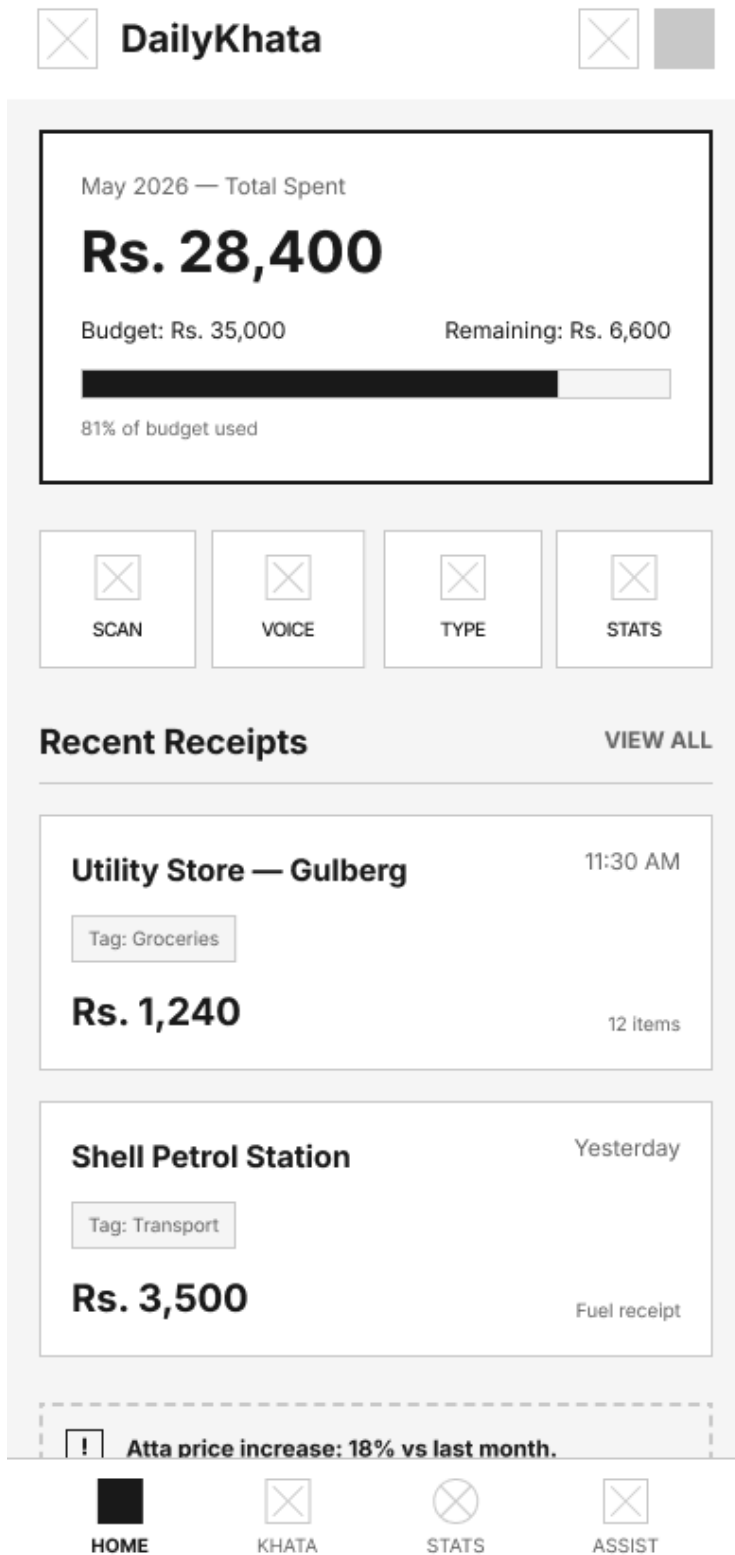


Figure 3: Med-fi Home Dashboard - action buttons, monthly spend card, recent entries and visible price alert.

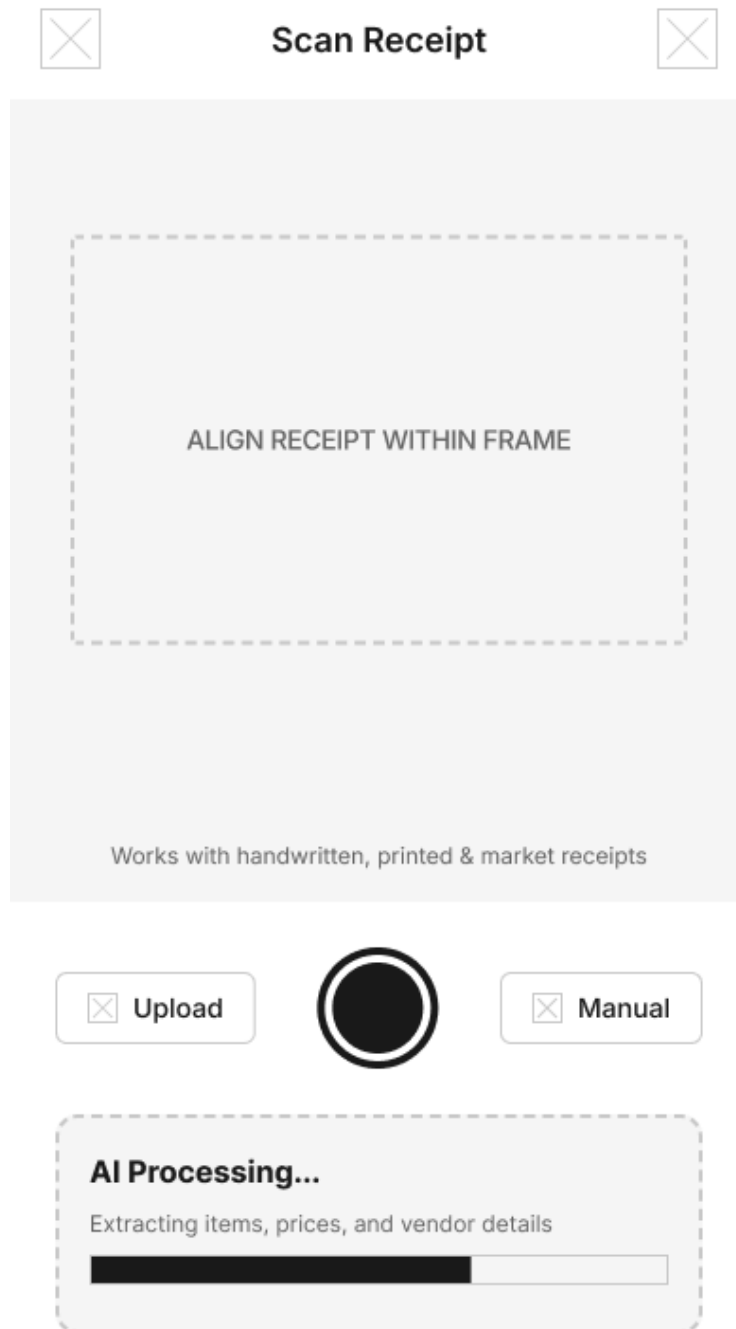


Figure 4: Med-fi Scan Receipt - camera frame, capture/gallery/manual options and AI processing state.



Verify Receipt

Save

Utility Store — Gulberg, Lahore

Receipt Date: 16 May 2026 · 11:30 AM

1000 · Food & Groceries

ITEM	QTY	UNIT	TOTAL
Atta (10kg)	1	212	Rs. 212
Doodh (1L)	4	180	Rs. 720
Chawal (5kg)	1	340	Rs. 340
Cooking Oil	2	480	Rs. 960
Total			Rs. 2,232

3 fields flagged — tap to correct before saving

Price Changes Since Last Visit

Atta: Rs. 180 → **Rs. 212** +18%

Doodh: Rs. 175 → **Rs. 180** +3%

CONFIRM & SAVE TO KHATA

EDIT ITEMS

Figure 5: Med-fi Receipt Verification - itemized table with stronger OCR correction prompt.

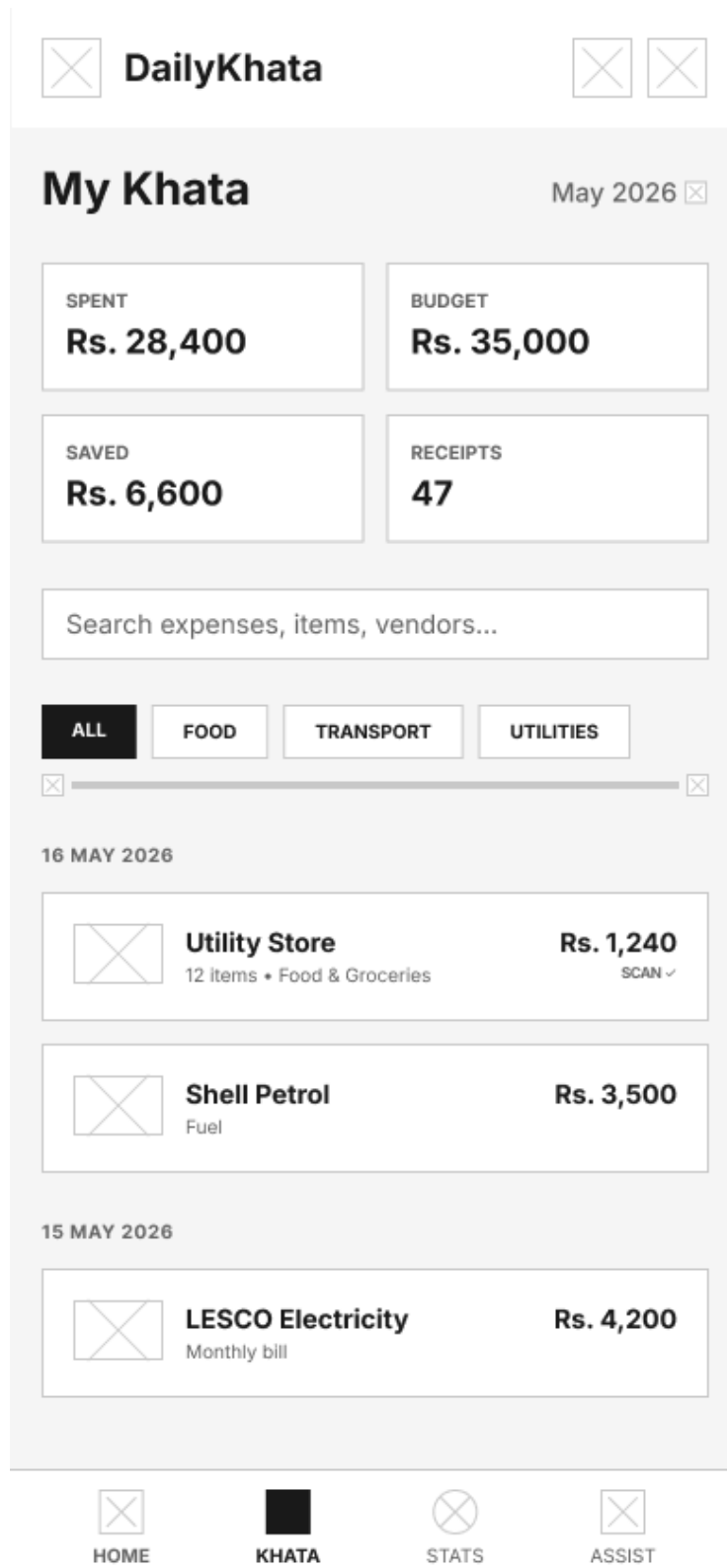


Figure 6: Med-fi Digital Khata - searchable ledger with local categories and date grouping.

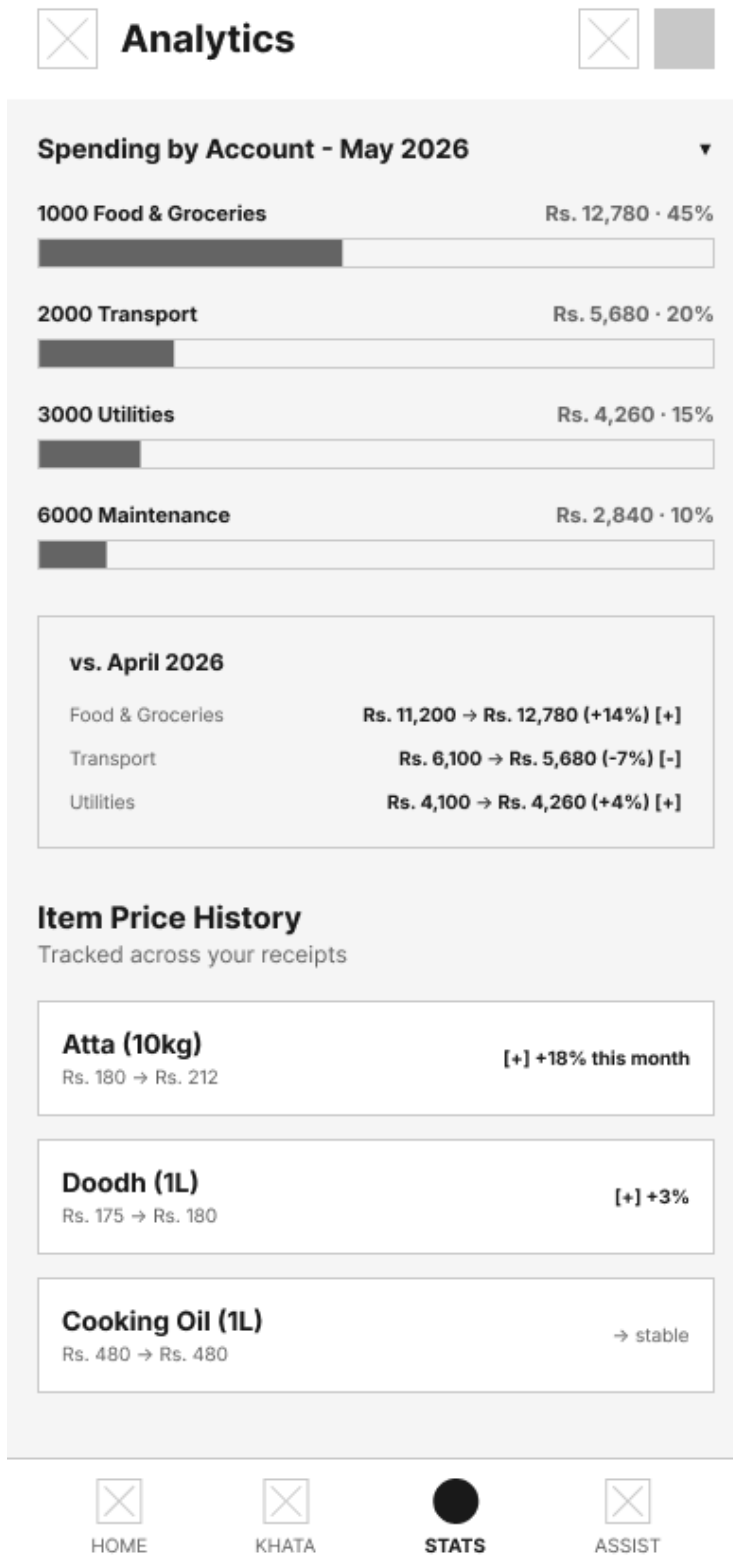


Figure 7: Med-fi Analytics - month carousel, category bars and May-versus-April insight card.

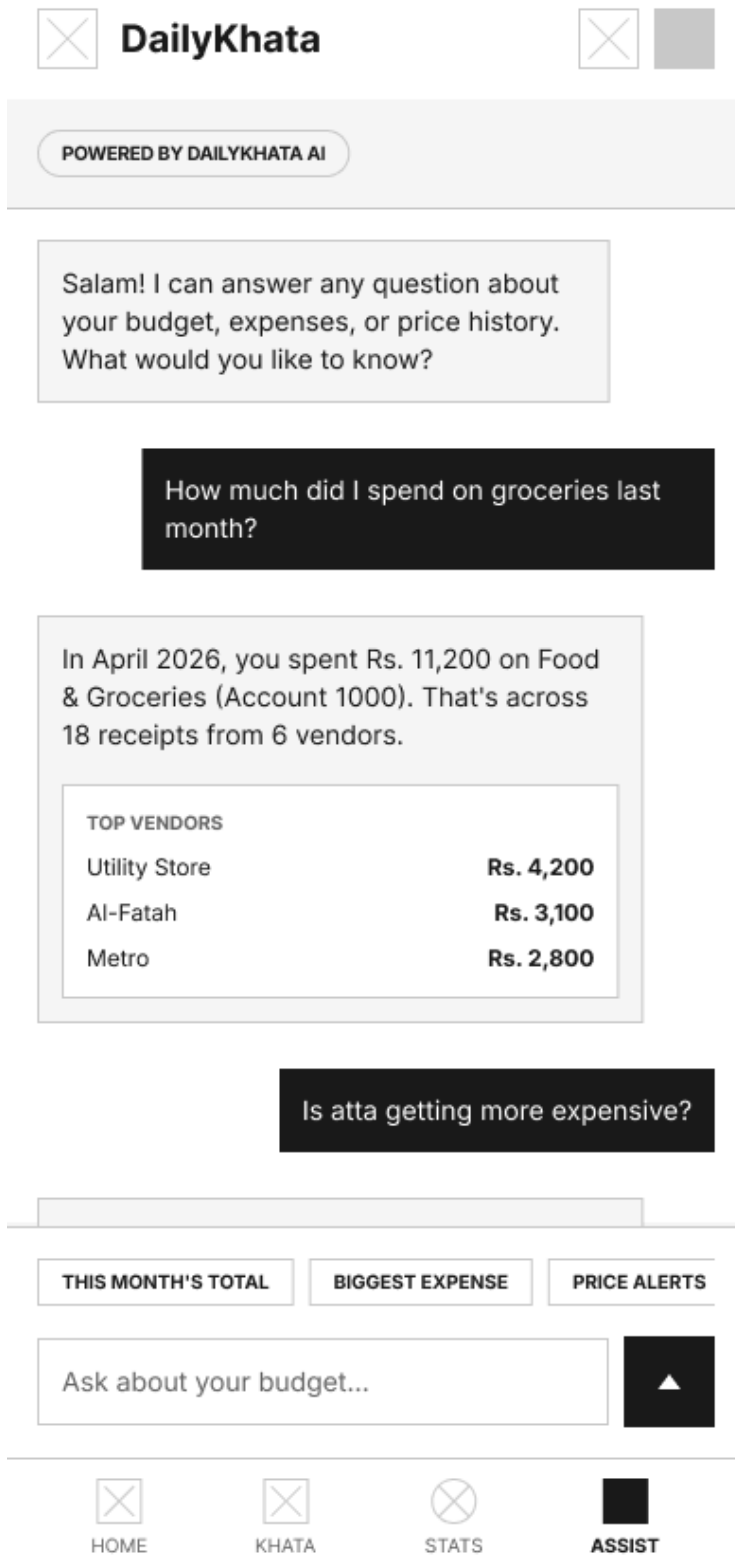


Figure 8: Med-fi AI Assistant - quick questions and plain-language answer card.

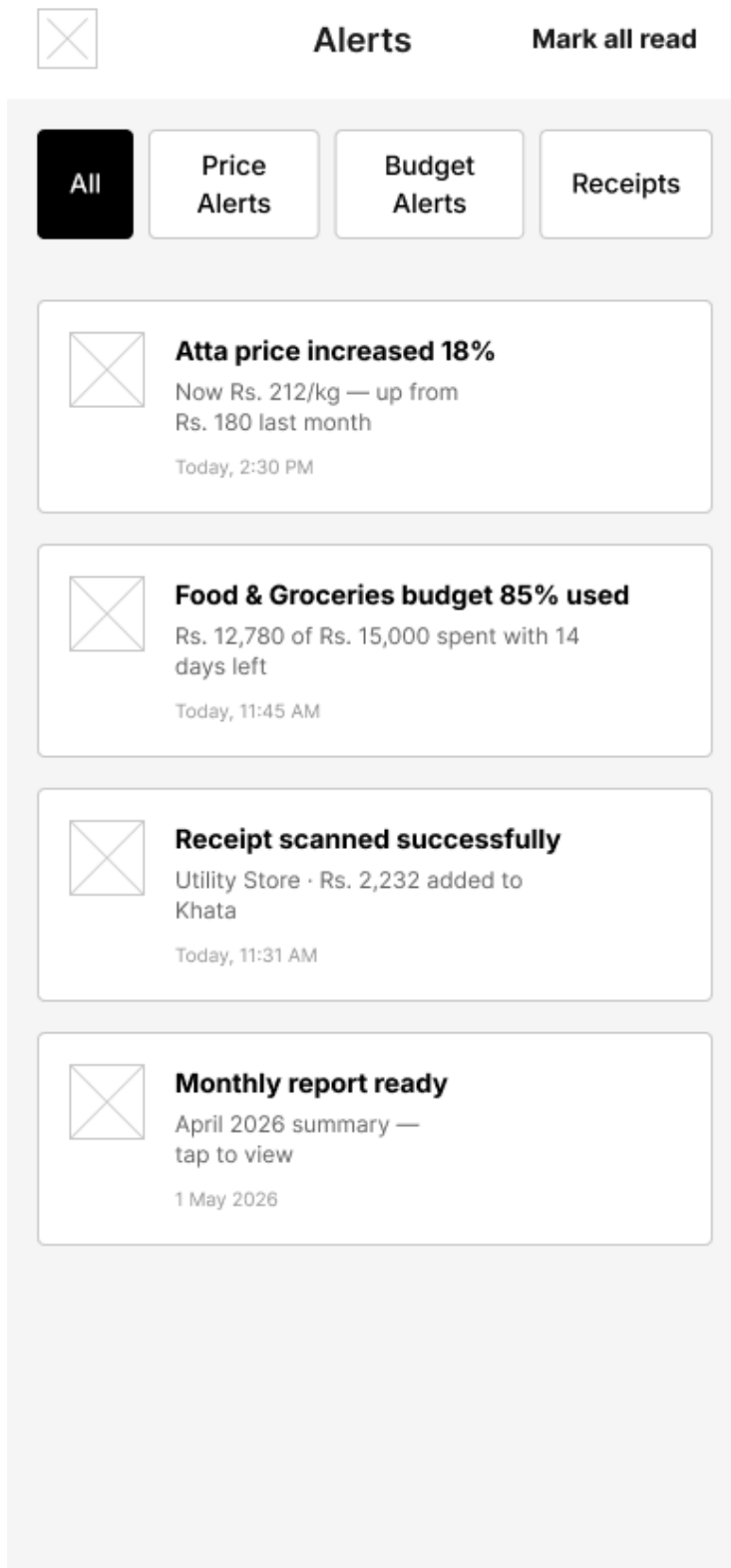


Figure 9: Med-fi Alerts - price and budget alerts with clear action pathway.

☐

Budget Settings

Save

MONTHLY INCOME

Rs. 75,000

Salary
Freelance
Both

Allocate Budget by Account

Following Chart of Accounts

1000 Food & Groceries	Rs. 18,000 24%
2000 Transport	Rs. 7,500 10%
4000 Healthcare	Rs. 3,750 5%
5000 Education	Rs. 5,250 7%
6000 Maintenance	Rs. 3,000 4%
Remaining / Savings	Rs. 31,500 42%

☐

Total allocated: Rs. 43,500 • Savings: Rs. 31,500

Figure 10: Med-fi Budget Setup - expanded Milestone 4 configuration task with local categories.

4.5 Core Navigation Flows

Milestone 4 required the prototype to widen task focus beyond the initial low-fidelity tasks. DailyKhata therefore retains the six tested tasks and adds two additional task flows: configuring a monthly budget and exporting/sharing a monthly summary. The main flows are listed below.

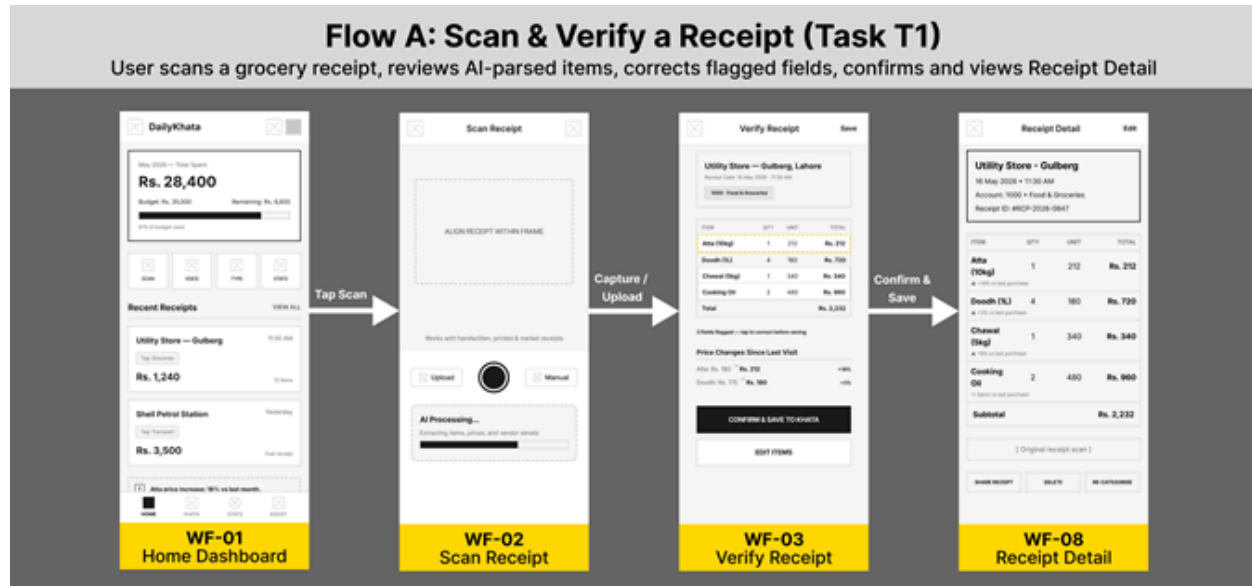


Figure 11: Core Flow A - scanning and verifying a receipt before saving to Khata.

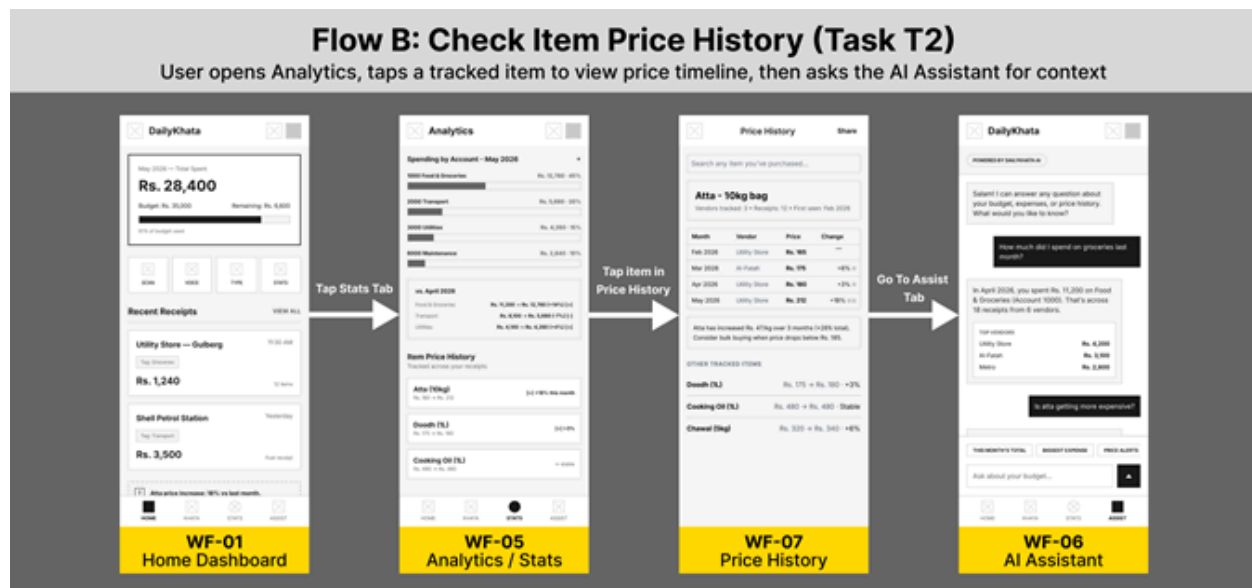


Figure 12: Core Flow B - responding to a price alert and using item history or AI insight.

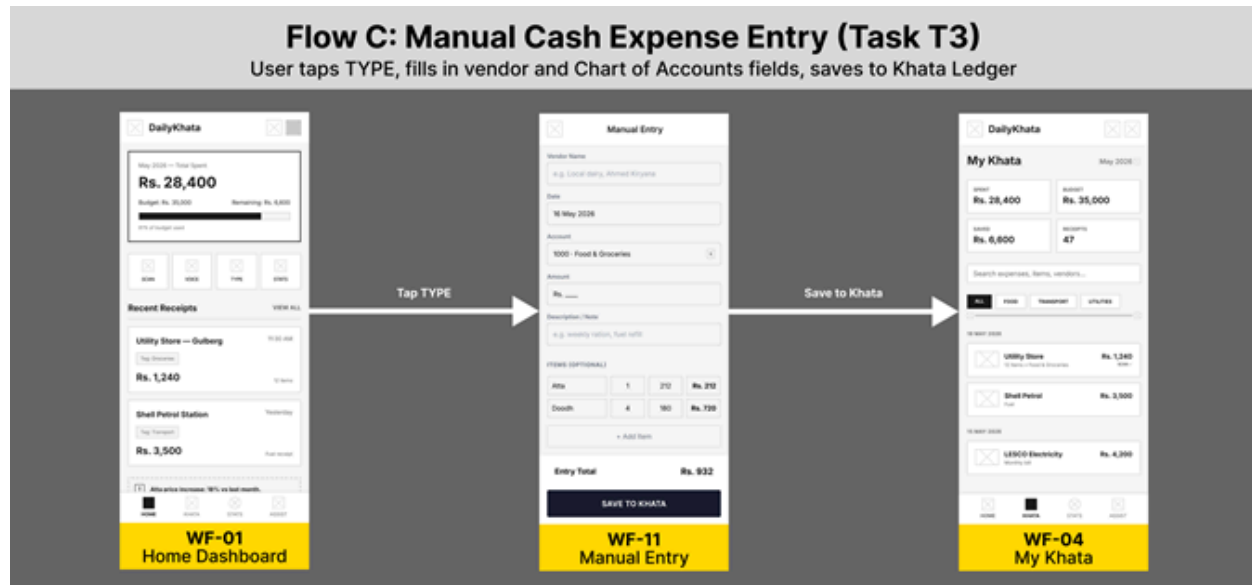


Figure 13: Core Flow C - configuring a monthly budget with local categories.

Flow	Screens Included	User Goal
A. Scan and Verify Receipt	Home -> Scan Receipt -> AI Processing -> Verify Receipt -> Receipt Detail	Convert physical receipt into accurate ledger entry.
B. Check Item Price History	Analytics -> Item Price History -> AI Assistant optional	Understand inflation trend for a specific item.
C. Manual Cash Expense	Home -> Manual Entry -> Khata Ledger	Capture no-receipt cash spending such as rickshaw or roadside purchase.
D. Ask AI Budget Question	Home/AI -> AI Assistant -> Data answer card	Get a plain-language summary without navigating charts.
E. Monthly Review	Khata -> Analytics -> Budget Setup	Compare month-to-month spending and adjust budget.
F. Price Alert Response	Home Alert -> Alerts -> Price History -> AI Assistant	Act on a meaningful price increase.
G. Budget Setup (new)	Home -> Budget Setup -> Category allocations -> Save	Configure monthly spending limits.
H. Export/Share Summary (new)	Analytics -> Summary -> Share PDF/WhatsApp	Share monthly summary with family or personal record.

4.6 Prototype Dummy Data

Data Area	Example Values Used	Purpose
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Receipts	Utility Store Gulberg, Metro Grocery, Shell Petrol Station, LESCO Electricity	Represents common Pakistani household expense sources.
Items	Atta 10kg, Doodh 1L, Chawal 5kg, Cooking Oil 1L, Petrol, Rickshaw Fare	Supports local price history and informal expense scenarios.
Budget	Monthly budget Rs. 35,000; May spend Rs. 28,400; remaining Rs. 6,600	Drives dashboard and analytics examples.
Price changes	Atta +18%, Cooking Oil +12%, Doodh +3%, Chawal +8%	Creates realistic alert and price-history tasks.
Categories	Ration/Groceries, Transport/Rickshaw, Bills/Utilities, Maintenance, Miscellaneous	Aligns with local mental models from needs finding.

4.7 Medium-Fidelity Validation Against Requirements

Milestone 4 Requirement	How DailyKhata Submission Addresses It
Conduct user test	Five moderated think-aloud sessions with participant profiles, tasks, timings and observations.
Analyze results	Task success table, SUS results, critical incidents, prioritized usability issues and UX findings.
Recommend solutions	Each issue is mapped to concrete design changes for the med-fi prototype.
Create med-fi prototype	Eight screen layouts, redesigned UI hierarchy and interactive flow descriptions are documented.
Widen task focus	Two additional flows added: Budget Setup and Export/Share Monthly Summary.
Choose scenarios and dummy data	Realistic Pakistani household data included for receipts, prices, categories and alerts.
No placeholder text in prototype	All screens use meaningful labels and realistic values; only this report marks image-replacement placeholders when actual screenshots are unavailable.
Share prototype link	Prototype link should be placed in the final presentation and Google Classroom private comment.

4.8 Prototype Link and Video Link Placeholders

Deliverable	Link / Status
Interactive med-fi prototype link	[Insert final Figma/Lovable prototype link here before submission]

Prototype walkthrough video link	[Insert final walkthrough video link here before submission]
Google Classroom private comment	Paste both links in the private comment as required by the Milestone 4 brief.

5. Risk Assessment

Risk	Likelihood	Impact	Mitigation
AI/OCR may still fail on heavily faded or mixed Urdu-English receipts	High	High	Keep mandatory verification step; allow manual correction; store original receipt image.
Low-literacy users may still find analytics overwhelming	Medium	Medium	Use plain-language summaries as default; keep charts secondary.
Offline-first functionality may complicate sync and data consistency	Medium	High	Queue entries locally and show sync status clearly.
Users may distrust financial data privacy	Medium	High	Add privacy explanation, local storage options and transparent account deletion/export flows.
Prototype lacks real OCR backend, so expectations may exceed functionality	Medium	Medium	Label prototype as simulated AI during testing; separate UI evaluation from backend feasibility.
Too many features may dilute simplicity	Medium	Medium	Preserve one-tap home actions and hide power-user tools behind secondary screens.

6. Conclusion and Next Steps

Milestone 4 demonstrates that DailyKhata has a strong and relevant concept for its intended Pakistani user base, but the low-fidelity prototype required meaningful usability improvements before moving forward. The user test confirmed the value of receipt scanning, price history and AI-based summaries, while also exposing important weaknesses in alert discoverability, OCR correction clarity, category terminology and analytics controls.

The medium-fidelity redesign directly addresses these problems through clearer visual hierarchy, stronger task cues, simplified language and expanded task coverage. The redesign remains

consistent with earlier Patches milestones: it continues to support multimodal input, mandatory verification, local spending categories, low-end Android usability and a bridge between traditional Khata habits and structured digital analytics.

Recommended next steps are to build the interactive Figma screens based on the layouts in this report, insert actual screenshots and links into the presentation, conduct a second usability test on the med-fi prototype, and compare the new SUS score and task success rates against the current low-fidelity baseline. A reasonable success target for the next iteration is an average SUS score above 75, T6 alert success above 80%, and near-zero missed OCR correction prompts during receipt verification.

Appendices

Appendix A: Moderator Script Summary

- Welcome participant and explain that the prototype is being tested, not the participant.
- Collect consent for participation and note taking/screen recording.
- Ask the participant to think aloud while completing each task.
- Read one task at a time without explaining where to tap.
- Record success, time, misclicks, critical incidents and notable quotes.
- Administer SUS questionnaire and open-ended debrief questions.

Appendix B: SUS Questionnaire Items

No.	SUS Item
1	I think that I would like to use this system frequently.
2	I found the system unnecessarily complex.
3	I thought the system was easy to use.
4	I think that I would need the support of a technical person to be able to use this system.
5	I found the various functions in this system were well integrated.
6	I thought there was too much inconsistency in this system.
7	I would imagine that most people would learn to use this system very quickly.
8	I found the system very cumbersome to use.
9	I felt very confident using the system.
10	I needed to learn a lot of things before I could get going with this system.

Appendix C: Raw Observation Log Summary

Participant	Most Positive Comment	Most Important Difficulty	Suggested Improvement
P01	"This can replace my notebook if the receipt saves correctly."	Could not find Alerts.	Show important warnings on home.
P02	"I like that it shows the items before saving."	Category code felt confusing.	Use normal household names for categories.
P03	"Price change is useful because prices keep increasing."	Confirm button was not visible enough.	Make save action larger and clearer.
P04	"AI questions are fast."	Wanted more comparative context in Analytics.	Add short explanation under charts.
P05	"This is better than keeping photos in Notion."	Expected auto-save after scan.	Explain why review is required before saving.